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$$\begin{aligned} I &= \int x^3 \cos x dx \\ \begin{cases} u = & x^3 \\ dv = & \cos x dx \end{cases} \Rightarrow \begin{cases} du = & 3x^2 dx \\ v = & \sin x \end{cases} \\ I &= x^3 \sin x - \int 3x^2 \sin x dx \\ &= x^3 \sin x - 3 \int x^2 \sin x dx \\ \begin{cases} u = & x^2 \\ dv = & \sin x dx \end{cases} \Rightarrow \begin{cases} du = & 2x dx \\ v = & -\cos x \end{cases} \\ I &= x^3 \sin x - 3 \left(-x^2 \cos x - \int -2x \cos x dx \right) \\ &= x^3 \sin x + 3x^2 \cos x - 6 \int x \cos x dx \\ \begin{cases} u = & x \\ dv = & \cos x dx \end{cases} \Rightarrow \begin{cases} du = & dx \\ v = & \sin x \end{cases} \\ I &= x^3 \sin x + 3x^2 \cos x - 6 \left(x \sin x - \int \sin x dx \right) \\ &= x^3 \sin x + 3x^2 \cos x - 6x \sin x + 6 \int \sin x dx \\ &= x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + c \\ &= (x^3 - 6x) \sin x + (3x^2 - 6) \cos x + c \end{aligned}$$

References